Central Bank Communication in Time of Crisis: Textual Complexity and Clarity Trade-off

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Abstract

Central banks may rationally adjust their communication strategies to balance clarity and flexibility, especially amid shifting economic conditions. Textual complexity can signal deliberate ambiguity, preserving policy maneuverability while still guiding expectations. This paper uses natural language tools to analyze policy statements from multiple central banks and their relationship to linguistic complexity, examining not only readability but also abstractness, informativeness, and disunity. We find that most dimensions of linguistic complexity increase during periods of low growth or economic stimulus, and abstract language is used more intensively during stimulation regardless of growth levels. By tracing these patterns, this study provides evidence that the degree of complexity of communications is the outcome of a rational choice made to balance competing central bank goals.

Keywords:

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1 Introduction

Central banks play a central role in stabilizing the economy through monetary policy, and communication has become a key tool in shaping expectations. Policy statements influence the decisions of financial institutions, households, and firms, making the clarity and structure of these texts an important component of policy transmission. Nevertheless, although clear signals to such economic agents can greatly encourage actions aligned with central bank goals, clearly stated plans of action can restrict the central bank's freedom of action in the future.

While many studies examine how central bank communication affects markets, far less attention has been paid to the ways in which economic conditions shape the communication itself—particularly its linguistic complexity. Historically, communication styles have often reflected policymakers' preferences. Greenspan's well-known remark ("I guess I should warn you, if I turn out to be particularly clear, you've probably misunderstood what I've said") illustrates a philosophy of managing expectations by avoiding excessive concreteness in communication. Over time, however, transparency pressures increased, as illustrated by the practice of releasing FOMC transcripts, beginning in 1993. More recently, initiatives like "Fed Listens" and the ECB's explicit goal of reducing complexity highlight a shift toward clearer communication, especially during crises when central banks aim to reassure the public.

Yet clarity carries risks: overly explicit commitments can restrict policy flexibility and expose central banks to credibility losses if conditions change. The ECB's 2011 rate-hike episode illustrates the potential for clear forward guidance to backfire when economic conditions deteriorate. This tension suggests that communication strategies may shift with the economic environment, particularly in times of stress.

This paper examines the evolution of linguistic complexity in central bank policy statements in tandem with economic conditions, focusing on the trade-off between clarity and flexibility. We analyze statements from major central banks and inflation-targeting regimes worldwide, moving beyond conventional metrics such as word count or readability. Our measures capture multiple dimensions of complexity—including abstractness, informativeness, and disunity (textual cohesion)—to identify international patterns in communication during crises and other extreme events

Our main question is whether textual complexity systematically co-moves with economic conditions—particularly during periods of acute stress. Complexity may rise or fall depending on whether central banks prioritize clarity or flexibility, and simple metrics often obscure these dynamics. By distinguishing between different dimensions, which capture distinct yet overlapping elements of complexity, we can identify which aspects shift during crises and how they shift. We additionally explore cross-country heterogeneity, audience-specific communication, and thematic shifts over time. These analyses complement the main results and are presented in Appendix E.

The results indicate substantial variation in how complexity metrics respond to policy rates and macroeconomic variables. Notably, linguistic complexity in central bank statements appears to correlate more strongly with periods of extremely low economic growth than with conventional indicators of downturns. In many cases, these statements become more difficult to read, convey

greater informational content, and exhibit reduced cohesion across sentence transitions.[1] Furthermore, irrespective of growth levels, when there is a need to stimulate the economy, statements tend to adopt more abstract language as interest rates decline. These results complement recent work using large language models to assess semantic clarity and explanatory depth in central bank communication, which finds increased complexity during periods of uncertainty (Silva et al., 2025). We also observe persistent regional differences in communication style, specifically with geographic location—specifically, between Eastern and Western countries—than with linguistic factors (e.g., English vs. non-English-speaking nations).

Related Literature. This paper contributes to the growing literature on central bank communication, particularly studies that use textual analysis to examine how communication characteristics relate to macroeconomic conditions and policy decisions.

The first major branch is related to **the effects of central bank communication on the economy.** Recent advances in computational linguistics have enabled researchers to analyze central bank communication at scale, shifting from manual coding to automated textual methods. Much of this work focuses on sentiment analysis and hawkish/dovish tone. Sentiment measures typically capture positivity, negativity, or dissent, with positive tone often associated with optimistic economic conditions and negative tone linked to caution or uncertainty. These measures have been used to study relationships between tone and policy uncertainty, volatility (Baranowski et al., 2021; Tillmann and Walter, 2019; Gu et al., 2022), unemployment, output growth, inflation (Bohl et al., 2023; Clements and Reade, 2020; Eugster and Uhl, 2024), interest rates (Astuti et al.), and financial markets (Picault and Renault, 2017; Ehrmann and Talmi, 2020; Schmeling and Wagner, 2023; Hayo et al., 2022). Although dissent provides insight into internal policy disagreements, it falls outside the scope of this paper, as we treat each central bank as a unified institution.

A related strand of research examines hawkishness and dovishness. Hawks prioritize inflation control even at the cost of slower growth, while doves emphasize employment and economic activity. These orientations serve as important indicators of policy stance. For example, Parle (2022) finds that a more hawkish tone in ECB press conferences raises stock prices, while Dossani (2021) shows that hawkish communication reduces the variance risk premium and increases option-implied risk aversion across several major central banks.

Compared with sentiment and tone, ambiguity and linguistic complexity have received less attention. Readability is the most common proxy for complexity, with the Flesch–Kincaid score widely used to assess accessibility. Readability has meaningful economic implications: opaque statements can hinder public understanding and force firms to rely on external interpretations, potentially increasing uncertainty. Readability scores have risen over time for both the Federal Reserve and the ECB (T. Hughes and Kesting, 2014), though the ECB has recently implemented reforms to simplify its communication. Readability has been linked to trading volume and volatility (Smales and Apergis, 2017), market reactions under unconventional monetary policy (Hayo et al.,

[1] These findings somewhat resonate with the theoretical framework of Cukierman and Meltzer (1986), which suggests that central banks may exploit surprise inflation—often facilitated by complex or vague communication—to stimulate output, particularly when such stimulus is most needed, as during a recession.

2022), and interactions with sentiment (Celler).

Beyond major central banks, several studies examine smaller institutions, though typically on a country-by-country basis (Binette and Tchebotarev, 2019; Doan et al., 2023). Cross-country analyses remain limited. Born et al. (2014) study Financial Stability Reports from 37 countries, focusing on positivity and financial market outcomes. In contrast, our work centers on linguistic complexity and its relationship with macroeconomic variables such as growth and inflation, particularly during recessions.

The study most closely aligned with ours is Luangaram and Wongwachara (2017), who analyze monetary policy statements from 22 central banks using topic modeling, sentiment, readability, and text similarity. However, their dataset ends in 2015 and does not include the Effective Lower Bound (ELB) period or the COVID-19 pandemic. Their methods also rely on standard ambiguity metrics, leaving room for more nuanced measures of complexity.

A recent contribution by the IMF (Silva et al., 2025) uses large language models to classify central bank statements by semantic clarity and explanatory depth, finding that complexity increases during periods of uncertainty. While their focus is on communicative intent and audience segmentation, our approach emphasizes linguistic structure—readability, abstractness, informativeness, and cohesion—providing a complementary perspective on how complexity responds to economic conditions, especially crises.

The second major branch relates to **target audiences and speaker heterogeneity**. Effective communication requires that economic agents understand monetary policy goals and strategies. While financial market participants and professional forecasters typically possess the expertise to interpret policy statements, the general public may lack the motivation or background knowledge to do so (van der Cruijsen et al., 2015). Public engagement tends to rise during major events such as the global financial crisis or the COVID-19 pandemic (Gros and Capolongo, 2020), and transparency initiatives can improve understanding among non-experts (Dräger et al., 2016).

A substantial theoretical literature highlights audience heterogeneity. Households, firms, and financial markets form expectations differently (Binder, 2017; Coibion and Gorodnichenko, 2015), suggesting that central banks may tailor communication to specific groups. Empirical work on this dimension remains limited. We contribute by applying the model of Pfeifer and Marohl (2023) to infer the intended audience of each statement, allowing us to explore how complexity varies across target groups.

Heterogeneity may also arise from the speakers themselves. Bennani and Neuenkirch (2016) shows that central bankers adjust their language depending on whether they address domestic or international audiences or speak before or after Governing Council meetings. In contrast, Jansen and de Haan (2013) finds that ECB communication has remained broadly consistent over time. To account for such variation, our empirical strategy employs panel fixed effects to control for unobserved heterogeneity across institutions.

Additional related literature and an extended discussion of audience-specific communication appear in Appendix A.

2 Data

This section describes the primary datasets used in the analysis: a corpus of central bank policy statements and a set of country-specific macroeconomic variables.

2.1 Central Bank Statements

The textual data consists of English-language policy statements collected from the public websites of 18 central banks. The sample includes three major institutions—the Federal Reserve (Fed), the European Central Bank (ECB), and the Bank of Japan (BoJ)—as well as a diverse set of central banks operating under inflation-targeting regimes across both advanced and emerging market economies.

The dataset covers the period from 2000 to 2025, spanning several pivotal episodes, including the pre- and post-Great Financial Crisis (GFC), effective lower bound (ELB) periods (where applicable), and the pre- and post-COVID-19 pandemic eras. Statement availability varies by country (see Table 3 in Appendix B). The initial dataset comprises 3,809 individual statements, which are aggregated to the quarterly level to ensure temporal consistency and facilitate regression analysis, yielding 1,592 observations.

Given differences in naming conventions for "policy statements" across central banks, we consistently select the press-release version.[2] This choice promotes uniformity in both length and communicative intent, aligning with the paper's focus on communication directed toward the broader public, including households. During data collection, we remove extraneous elements such as dates, timestamps, signatures, and administrative notices, so that the textual analysis focuses on the substantive content of the statements.

2.2 Linguistic Complexity Measures

We construct four main dimensions of textual complexity from the statements: readability, abstractness, informativeness, and disunity. Simplified examples, detailed implementation, and full metric definitions appear in Appendix B. The hypotheses linking these measures to economic conditions are presented in Section 3 (Theory and Empirical Framework).

Readability captures the structural difficulty of a text, reflecting sentence length, word complexity, and other surface-level linguistic features. We compute several standard indices—Gunning–Fog, SMOG, Flesch Reading Ease, Flesch–Kincaid Grade Level, Automated Readability Index (ARI), Coleman–Liau, LIX, and RIX. Given the high correlation among these measures, we apply Principal Component Analysis (PCA) and retain the first principal component as a composite readability index, interpreted so that higher values indicate greater textual complexity.[3]

[2] For the ECB, we use the introductory statement from its press conferences, which is (i) more comparable to other public-facing communications, (ii) more informative in earlier periods, and (iii) emphasized by the ECB as a primary communication channel (Bernoth and Dany-Knedlik, 2020).

[3] Bartlett's test yields a p-value of 0, and pairwise correlations range from 0.73 to 0.98, with most exceeding 0.90. Similar patterns hold for the other complexity dimensions.

Abstractness is constructed as the inverse of concreteness, reflecting the extent to which texts rely on intangible or conceptually demanding language. Word-level concreteness scores are drawn from three established sources: the Brysbaert dataset (Brysbaert et al., 2013), the Glasgow Norms (Scott et al., 2019), and the MRC Psycholinguistic Database (Coltheart, 1981), with values typically ranging from 1 (most abstract) to 5 (most concrete). We compute document-level concreteness using means and concrete/abstract ratios, invert these measures to obtain abstractness, and summarize the resulting variants using PCA. The first principal component serves as our abstractness index, with higher values indicating greater complexity.

Informativeness is measured using Shannon entropy, capturing token diversity and vocabulary richness. Higher entropy indicates more diverse vocabulary and greater informational content, which we treat as a distinct dimension of complexity. We normalize entropy before including it in the analysis with other complexity measures.

Disunity captures the lack of semantic cohesion between sentences, based on the cosine similarity of sentence-level word embeddings. We compute first-order coherence—adjacent sentences (Bedi et al., 2015)— and second-order coherence—sentences separated by one intervening sentence (Parola et al., 2023). Higher similarity scores in this setting indicate lower cohesion and thus greater disunity. These measures are combined using PCA, and the first principal component is used as our disunity index, where higher values denote less cohesion and higher complexity.

Additional textual metrics—such as dependency distance, perplexity, and basic length measures—were considered but excluded from the main analysis due to interpretability and redundancy. Average complexity scores by country are reported in Table 5 in Appendix B. While complexity dimensions do not move uniformly across countries, higher overall complexity tends to be associated with higher readability scores, greater abstractness, increased informativeness, and elevated disunity.

2.3 Country Characteristics

To examine how textual complexity co-moves with economic conditions, we compile a set of macroeconomic variables for each country: real GDP growth, inflation, unemployment rate, policy rates, and recession indicators. Wherever possible, data span 2000–2024.

Real GDP data are primarily sourced from FRED, supplemented with national central bank data when quarterly series are unavailable. We use quarter-to-quarter real GDP growth as our baseline measure. To capture periods of acute economic stress, we construct a quantile-based recession indicator that indicates observations in the bottom 2nd percentile of each country's growth distribution; alternative thresholds (5th and 10th percentiles) are considered as robustness checks. We also incorporate conventional recession indicators: for non-U.S. economies, we use the OECD midpoint method, while for the United States, we additionally include the NBER recession indicator.

The **Consumer Price Index (CPI)** and **policy rate** data are obtained from the BIS. Inflation is computed as the quarterly growth rate of the CPI. For the policy rate, we use both the level and

its first difference to capture the direction and magnitude of monetary policy adjustments. We also construct a binary indicator for whether the policy rate changed in a given quarter; results using this measure appear in Appendix E.

The **unemployment rates** are drawn from the OECD, supplemented with national sources for countries not covered by the OECD database. We use the total unemployment rate for individuals aged 15 and above. Minor differences in age definitions across countries are retained to reflect official national statistics.

Table 6 in Appendix C reports summary statistics for all macroeconomic variables. The considerable heterogeneity across countries underscores the importance of controlling for country fixed effects in the empirical analysis. Additional details on data construction and alternative definitions appear in Appendix C.

3 Theory and Empirical Framework

Central bank communication faces a trade-off between clarity—anchoring expectations—and flexibility—preserving room for future adjustments. We outline the theoretical foundation and hypothesis of this trade-off, then develop the reduced form used in our regressions.

3.1 Theory and Hypothesis

There is a fundamental trade-off in central bank communication strategies. On one hand, clarity is essential for anchoring expectations and shaping the behaviour of households and firms. On the other hand, vagueness—approximated by complexity—preserves flexibility, allowing policymakers to adapt their narrative as economic conditions evolve. This tension between clarity and flexibility is crucial to understanding policy communication behaviours. Formally, the trade-off can be expressed as:

$$LC_{c,t}(anchoring) = f_a(\mu_c, i_{c,t}, \mathbf{X_{c,t}}).LC_{c,t}(flexibility) = f_f(\mu_c, i_{c,t}, \mathbf{X_{c,t}}). \tag{1}$$

where $LC_{c,t}$ denotes one of the four complexity dimensions (readability, abstractness, informativeness, disunity), $i_{c,t}$ is the policy rate, $X_{c,t}$ is the vector of macroeconomic variables, and μ_c captures country-specific unobserved heterogeneity. The overall linguistic complexity is modelled as the weighted average of the two forces, with weights α and $1-\alpha$, respectively, and $\alpha \in [0, 1]$:

$$LC_{c,t} = \alpha LC_{c,t}(flexibility) - (1-\alpha)LC_{c,t}(anchoring), \tag{2}$$

A higher weight α indicates a greater emphasis on flexibility and, correspondingly, less emphasis on clarity. Because of data limitations and the difficulty of identifying the structural trade-off directly, we rely on reduced-form relationships. Appendix D shows that linguistic complexity can be expressed as a function of policy decisions, economic conditions, and speaker-specific factors. Thus, in reduced form, the trade-off can be written as:

$$LC_{c,t} = \text{const} + A_1\mu_c + A_2 i_{c,t} + A_3 X_{c,t} + u_{c,t}, \tag{3}$$

Equation 3 can be estimated using ordinary least squares (OLS). To account for μ_c, we estimate the reduced form using panel fixed-effects models.

This framework motivates our empirical analysis. Specifically, we hypothesize that:

- Policy rate decisions (A_2) are linked to shifts in linguistic complexity, though the direction of the effect is theoretically uncertain. The sign likely depends on whether the central bank is focused on stimulus or inflation control, and on which dimension of complexity is under consideration.
- Economic conditions (A_3)—particularly GDP growth, inflation, and unemployment—systematically shape complexity. In extreme downturns, the ability to influence economic agents (e.g. to maintain consumption and investment levels) becomes of dominant importance. Such conditions therefore tend to increase clarity, informativeness and, because uncertainty tends to increase in unusually severe circumstances, disunity as well.

If valid, these hypotheses imply that communication clarity is not solely the result of predetermined preferences of central bank leaders, nor a random process, but a choice variable that central banks deliberately optimize in response to changing economic circumstances. Details of the theory, hypothesis, and derivations are presented in Appendix D.

3.2 Regression Specifications

We estimate two baseline fixed-effects models. The first uses a quantile-based recession indicator constructed from the bottom 2nd percentile of real GDP growth:

$$LC_{i,t} = \alpha_0 + \eta_i + \alpha_1 \Delta i_{i,t} + \alpha_2 g^{2p}_{y_{i,t}} + \alpha_3 g^{2p}_{y_{i,t-1}} + \alpha_6 \pi_{i,t} + \alpha_7 \pi_{i,t-1} + \alpha_8 u_{i,t} + \alpha_9 u_{i,t-1} + \epsilon_{i,t} \tag{4}$$

The second uses conventional recession indicators:

$$LC_{i,t} = \alpha_0 + \eta_i + \alpha_1 \Delta i_{i,t} + \alpha_4 RI_{i,t} + \alpha_5 RI_{i,t-1} + \alpha_6 \pi_{i,t} + \alpha_7 \pi_{i,t-1} + \alpha_8 u_{i,t} + \alpha_9 u_{i,t-1} + \epsilon_{i,t} \tag{5}$$

where LC indicating one of the four linguistic complexity measurements (readability, abstractness, informativeness and disunity), η_i as country fixed effect, Δi as policy rate change, g^{2p}_y as the bottom 2 percentile of the real GDP growth, RI as recession indicators, π as inflation rate and u as unemployment rate. Note that we use a threshold variable and a recession indicator to represent unusually severe economic conditions. The lagged values are added for all economic condition variables to ensure that countries that do not observe the exact economic conditions at time t can still observe them at time $t-1$. It is also necessary to include the inertia in considering past economic

conditions in decision-making at time t. We also estimate these two baseline fixed-effects models for geographic and linguistic subgroups. Full subgroup definitions, additional estimates, details, derivations, and robustness checks appear in Appendix D.

4 Results and Discussion

Before turning to the empirical results, we briefly summarize the hypotheses for the key coefficients. The coefficient on the change in the policy rate is expected to be statistically significant, though the direction is theoretically ambiguous and may differ across complexity dimensions. By contrast, coefficients on recession indicators — and to some extent inflation and unemployment — are expected to be positive across complexity measures.

4 5

Tables 1 and 2 report the baseline fixed-effects estimates for the four dimensions of linguistic complexity. Model 1 uses the quantile recession indicator (bottom 2nd percentile of real GDP growth), while Model 2 uses the 3D recession indicator. Results are shown for the full sample and for geographic and linguistic subgroups.

Readability. A clear distinction emerges between the two recession measures. In Model 1, extremely low GDP growth is strongly associated with higher readability-based complexity. The quantile recession indicator is positive and statistically significant across the full sample and Western countries (1% level), and for both linguistic subgroups (5% level). Lagged values show similar patterns. By contrast, the 3D recession indicator in Model 2 exhibits no significant relationship with readability.

For Eastern countries, policy rate changes play a more prominent role: decreases in the policy rate are associated with higher complexity, suggesting more nuanced communication during accommodative phases. Unemployment also matters. In Model 1, higher unemployment is linked to increased reading difficulty for the full sample, Western countries, Eastern countries, and English-speaking countries. Model 2 shows weaker but directionally similar effects.

Abstractness. Abstractness shows a clear pattern: across the full sample and most subgroups, policy rate decreases are linked to higher abstractness, indicating that monetary easing is accompanied by more conceptual or less concrete language. The effect is strongest in English-speaking countries, where the magnitude is more than twice that observed in non-English-speaking economies. Eastern countries show the same direction but lack statistical significance, likely due to a smaller sample size.

Lagged inflation also matters in Model 2: higher inflation in the previous quarter predicts greater abstractness for the full sample, Western countries, and English-speaking countries.

Informativeness. Informativeness responds most consistently to extreme GDP contractions. The quantile recession indicator is positive and significant for the full sample and both geographic subgroups, including lagged values for Eastern countries. This suggests that extremely low growth prompts central banks to provide more information. No significant effects appear for linguistic

Table 1: **Fixed effect models for all and sub-groups of countries (part 1)**

	All		Geography				Language			
			Western		Eastern		English speaking		Non-English speaking	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
	A. Readability									
Δi_t	−0.013	0.048	−0.024	−0.012	0.657*	0.562	0.115	−0.411	−0.085	0.042
	(0.101)	(0.104)	(0.113)	(0.142)	(0.360)	(0.390)	(0.136)	(0.286)	(0.118)	(0.078)
$g^{2p}_{y_t}$	1.312***		1.493***		0.824		1.184**		1.351**	
	(0.400)		(0.453)		(0.928)		(0.468)		(0.582)	
$g^{2p}_{y_{t-1}}$	0.627**		0.709**		0.206		0.762		0.445	
	(0.300)		(0.356)		(0.684)		(0.782)		(0.325)	
RI_t		0.043		−0.373		0.693		−0.618		0.376
		(0.342)		(0.360)		(0.590)		(0.644)		(0.385)
RI_{t-1}		0.199		−0.136		0.104		−0.876		0.541**
		(0.304)		(0.368)		(0.323)		(0.559)		(0.272)
π_t	0.008	−0.066	0.078	0.012	−0.011	−0.078	−0.028	−0.222	0.042	−0.017
	(0.061)	(0.081)	(0.070)	(0.109)	(0.143)	(0.131)	(0.151)	(0.172)	(0.059)	(0.089)
π_{t-1}	−0.057	0.020	−0.068	0.005	0.104	0.163	−0.034	0.209***	−0.060	0.011
	(0.045)	(0.071)	(0.045)	(0.055)	(0.216)	(0.226)	(0.108)	(0.080)	(0.051)	(0.088)
u_t	−0.053	0.042	0.038	0.226*	0.083	0.116***	0.062	0.263*	−0.080	−0.012
	(0.070)	(0.097)	(0.069)	(0.118)	(0.055)	(0.021)	(0.121)	(0.159)	(0.071)	(0.085)
u_{t-1}	0.140*	0.048	0.224**	0.033	0.078**	0.036	0.313**	0.145	0.111	0.055
	(0.083)	(0.088)	(0.104)	(0.141)	(0.037)	(0.037)	(0.150)	(0.178)	(0.099)	(0.084)
R^2	0.02	0.01	0.11	0.11	0.06	0.08	0.1	0.14	0.01	0.03
N	1323	1192	1043	943	280	249	479	423	844	769
	B. Abstractness									
Δi_t	−0.301***	−0.266***	−0.259***	−0.286***	−1.093	−0.565	−0.560***	−0.496***	−0.250**	−0.247***
	(0.098)	(0.090)	(0.084)	(0.092)	(0.898)	(1.016)	(0.090)	(0.139)	(0.098)	(0.084)
$g^{2p}_{y_t}$	0.476		0.327		1.219		−0.043		0.731	
	(0.393)		(0.417)		(1.326)		(0.409)		(0.550)	
$g^{2p}_{y_{t-1}}$	0.089		−0.001		0.938		0.053		0.140	
	(0.248)		(0.298)		(0.794)		(0.255)		(0.343)	
RI_t		−0.030		0.001		0.016		0.128		−0.121
		(0.178)		(0.146)		(0.479)		(0.146)		(0.251)
RI_{t-1}		0.351		−0.092		1.479*		−0.056		0.577
		(0.290)		(0.112)		(0.864)		(0.089)		(0.433)
π_t	0.047	0.048	0.037	0.041	0.195	0.139	−0.029	−0.064	0.059	0.094
	(0.044)	(0.055)	(0.048)	(0.073)	(0.156)	(0.108)	(0.058)	(0.060)	(0.049)	(0.066)
π_{t-1}	0.087	0.127*	0.063	0.101**	0.249	0.250*	0.111*	0.221**	0.093	0.097
	(0.054)	(0.067)	(0.044)	(0.048)	(0.186)	(0.146)	(0.058)	(0.086)	(0.066)	(0.076)
u_t	0.083	0.102*	0.095	0.143**	−0.081	−0.024	0.073	0.077	0.057	0.084
	(0.064)	(0.060)	(0.074)	(0.061)	(0.247)	(0.202)	(0.140)	(0.106)	(0.064)	(0.059)
u_{t-1}	0.086	0.074	0.084	0.049	−0.041	−0.046	0.056	0.070	0.112	0.094
	(0.076)	(0.061)	(0.103)	(0.072)	(0.131)	(0.130)	(0.112)	(0.080)	(0.093)	(0.075)
R^2	0.1	0.12	0.11	0.14	0.09	0.15	0.04	0.05	0.12	0.15
N	1332	1200	1043	943	289	257	479	423	853	777

Note: This table reports coefficient estimates from regressions 4 and 5, referred to as Model 1 and Model 2, respectively. Columns (1)–(2) use the full sample; (3)–(4) restrict to Western countries; (5)–(6) to Eastern countries; (7)–(8) to English-speaking countries; and (9)–(10) to non-English-speaking countries. Panels A–D report results using each complexity measure as the dependent variable, while the left-hand side lists all explanatory variables. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The number of observations and R^2_{adj} for each specification appear at the bottom of each panel.

Table 2: **Fixed effect models for all and sub-groups of countries (part 2)**

	All		Geography				Language			
			Western		Eastern		English speaking		Non-English speaking	
	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2	Model 1	Model 2
	C. Informativeness									
Δi_t	0.092	0.093	0.079	0.083	0.194*	0.073	0.103**	0.039	0.017	0.042
	(0.078)	(0.065)	(0.079)	(0.065)	(0.107)	(0.045)	(0.040)	(0.070)	(0.042)	(0.043)
$g^{2p}_{y_t}$	0.273**		0.286*		0.407**		0.431		0.224	
	(0.123)		(0.149)		(0.189)		(0.272)		(0.151)	
$g^{2p}_{y_{t-1}}$	0.135		0.145		0.241**		0.146		0.098	
	(0.102)		(0.128)		(0.096)		(0.323)		(0.132)	
RI_t		−0.062		−0.007		−0.146***		0.153		−0.149*
		(0.087)		(0.112)		(0.049)		(0.166)		(0.082)
RI_{t-1}		−0.032		−0.090		0.127		−0.210***		−0.060
		(0.070)		(0.083)		(0.092)		(0.047)		(0.099)
π_t	−0.015	0.036	−0.008	0.044	−0.027	0.003	0.081**	0.113*	0.003	0.037
	(0.031)	(0.035)	(0.032)	(0.039)	(0.044)	(0.030)	(0.040)	(0.068)	(0.027)	(0.040)
π_{t-1}	−0.001	−0.110**	0.003	−0.112**	−0.021	−0.085**	0.039**	−0.047	0.008	−0.069**
	(0.012)	(0.045)	(0.012)	(0.055)	(0.030)	(0.035)	(0.018)	(0.048)	(0.014)	(0.032)
u_t	0.003	0.049	−0.023	0.044	−0.034*	−0.013	0.152**	0.208**	−0.042	−0.001
	(0.052)	(0.047)	(0.064)	(0.058)	(0.018)	(0.008)	(0.062)	(0.102)	(0.039)	(0.017)
u_{t-1}	0.057**	0.024	0.089***	0.029	−0.031***	−0.021***	0.215***	0.158***	0.046	0.017
	(0.026)	(0.015)	(0.032)	(0.023)	(0.006)	(0.008)	(0.058)	(0.026)	(0.045)	(0.031)
R^2	0.05	0.09	0.04	0.08	0.17	0.21	0.41	0.42	0.01	0.04
N	1332	1200	1043	943	289	257	479	423	853	777
	D. Disunity									
Δi_t	0.031	0.079	0.048	0.111	−0.195	−0.329	0.052	0.049	−0.018	0.012
	(0.082)	(0.085)	(0.082)	(0.089)	(0.456)	(0.445)	(0.076)	(0.091)	(0.097)	(0.085)
$g^{2p}_{y_t}$	0.454**		0.697***		−0.296		0.369		0.484*	
	(0.195)		(0.212)		(0.218)		(0.227)		(0.253)	
$g^{2p}_{y_{t-1}}$	−0.023		0.128		−0.144		−0.146		−0.012	
	(0.150)		(0.154)		(0.292)		(0.232)		(0.206)	
RI_t		0.156		0.072		0.316		0.126		0.192
		(0.163)		(0.174)		(0.349)		(0.089)		(0.252)
RI_{t-1}		0.147		0.186		−0.236		0.029		0.135
		(0.169)		(0.172)		(0.280)		(0.026)		(0.234)
π_t	0.058	0.066	0.051*	0.063	0.149	0.150	0.002	−0.036	0.098**	0.124**
	(0.037)	(0.047)	(0.030)	(0.045)	(0.105)	(0.122)	(0.053)	(0.056)	(0.044)	(0.057)
π_{t-1}	−0.039	−0.063	−0.062**	−0.115**	0.152*	0.184*	−0.026	−0.024	−0.040	−0.061
	(0.030)	(0.053)	(0.030)	(0.053)	(0.089)	(0.107)	(0.029)	(0.050)	(0.036)	(0.066)
u_t	−0.037	−0.009	−0.054	0.004	0.020	−0.011	0.110***	0.113***	−0.103*	−0.073
	(0.048)	(0.047)	(0.067)	(0.063)	(0.039)	(0.045)	(0.030)	(0.025)	(0.056)	(0.055)
u_{t-1}	0.088**	0.069**	0.123**	0.073*	−0.029	−0.026	0.088***	0.076**	0.116*	0.095*
	(0.038)	(0.028)	(0.048)	(0.039)	(0.038)	(0.047)	(0.030)	(0.032)	(0.064)	(0.053)
R^2	0.02	0.03	0.04	0.05	0.17	0.19	0.24	0.24	0.02	0.03
N	1323	1192	1043	943	280	249	479	423	844	769

Note: This table reports coefficient estimates from regressions 4 and 5, referred to as Model 1 and Model 2, respectively. Columns (1)–(2) use the full sample; (3)–(4) restrict to Western countries; (5)–(6) to Eastern countries; (7)–(8) to English-speaking countries; and (9)–(10) to non-English-speaking countries. Panels A–D report results using each complexity measure as the dependent variable, while the left-hand side lists all explanatory variables. Standard errors are reported in parentheses. *, **, and *** denote significance at the 10%, 5%, and 1% levels, respectively. The number of observations and R^2_{adj} for each specification appear at the bottom of each panel.

subgroups.

Policy rate changes also matter: larger increases in the policy rate are associated with higher informativeness for Eastern and English-speaking countries. Economic conditions play a central role. Lagged inflation is negatively associated with informativeness across most specifications, while lagged unemployment is positively associated with informativeness, often at the 1% level. Exceptions include non-English-speaking countries (no unemployment effect) and Eastern countries (inverted unemployment effect).

Interestingly, Eastern countries show conflicting recession signals: quantile recessions increase informativeness, while 3D recessions decrease it. Non-English-speaking countries show a similar but weaker pattern.

Disunity broadly mirrors the readability results. Extremely low GDP growth increases disunity in the full sample and most subgroups, while conventional recession indicators show no significant effects. English-speaking countries follow the same directional pattern but lack statistical significance.

Unemployment again plays a role: higher lagged unemployment increases disunity across most samples, except Eastern countries. For English-speaking countries, current unemployment also matters. Additional patterns include: (i) lower lagged inflation increases disunity in Western countries, and (ii) higher current inflation increases disunity in non-English-speaking countries.

Summary. Across all four dimensions—readability, abstractness, informativeness, and disunity—extremely low GDP growth and elevated unemployment (particularly lagged) are associated with higher textual complexity. Monetary easing is linked to greater abstractness, while inflation and unemployment shape informativeness and disunity. Conventional recession indicators generally lack predictive power.

Eastern countries exhibit the most inconsistent patterns, while non-English-speaking countries show weaker significance overall but are consistent with their English-speaking counterparts. This may reflect either greater linguistic uniformity or institutional divergence in communication practices.

5 Conclusions, Limitations and Future Directions

This study uncovers consistent patterns in the linguistic complexity of central bank communications during periods of economic distress. Across most samples, statements become harder to read, more information-dense, and less cohesive across sentence transitions when real GDP growth falls into extreme contraction. These patterns do not appear during conventional recession episodes or periods of exceptionally high inflation. Abstractness, meanwhile, tends to rise when monetary easing is required, conditional on broader macroeconomic conditions. Unemployment emerges as a robust predictor of complexity, with higher rates generally associated with more intricate language. These relationships, however, are not uniform: Eastern economies display mixed and sometimes contradictory signals, and substantial variation across individual countries highlights the impor-

tance of institutional context and speaker-specific characteristics.

Beyond macroeconomic triggers, the analysis shows that central banks devote considerable attention to the general public—including households and firms—underscoring the importance of audience-targeted communication. Central banks with more hawkish or optimistic tones tend to produce more complex statements, possibly reflecting the abstract nature of inflation-focused messaging. This is further supported by the inflation topic, which correlates positively with abstractness.

Several limitations remain. The dataset, based on Luangaram and Wongwachara (2017), could be expanded to include additional central banks to improve coverage and generalizability. The empirical framework is intentionally simple; future work could incorporate non-linear models or more sophisticated structures capable of disentangling the dual forces of anchoring and flexibility. Finally, while this study provides descriptive evidence on audience targeting and linguistic complexity, further theoretical and empirical work is needed to explore these dynamics more rigorously.

Overall, this paper deepens our understanding of how central banks adjust their communication strategies in response to economic conditions and introduces a multidimensional framework for measuring textual complexity. By moving beyond traditional readability metrics and incorporating abstractness, informativeness, and disunity, the analysis offers a richer perspective on the strategic use of language in monetary policy. This approach opens the door for future research on how complexity interacts with audience targeting, sentiment, and institutional context, and raises the possibility of developing quantitative guidelines for effective communication under varying economic conditions. As central banks navigate increasingly volatile environments, the clarity, tone, and structure of their communication will remain central to policy transmission, and this paper provides a foundation for understanding that complexity with greater nuance.